



# Modelling Financial Markets

## Chapter 19: Review and Project Presentations



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## Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CV: Cross-Validation
- ▣ DCC: Dynamic Conditional Correlation
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ EVT: Extreme Value Theory
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FX: Foreign eXchange
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation

## Acronyms used in this material (2/2)

- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ SML: Security Market Line
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index

## Outline

Introduction

Review I: Markets, Returns and Pricing

Review II: Volatility, Dependence and Risk

Review III: High Frequency, Continuous Time and Options

Review IV: Learning, Text and New Markets

Common Pitfalls

Integrated Case Study: BET

The Team Project

Using AI in Your Project

Project Seeds

Conclusions

References

## What This Chapter Does

### Purpose

- Review Chapters 0–18 as one connected story
- Collect the pitfalls that returned in every chapter
- Run one integrated case study, from raw prices to a backtested value-at-risk (VaR) at 1%
- Prepare the team project: its four steps, the question, the methods and the AI policy

### Assessment

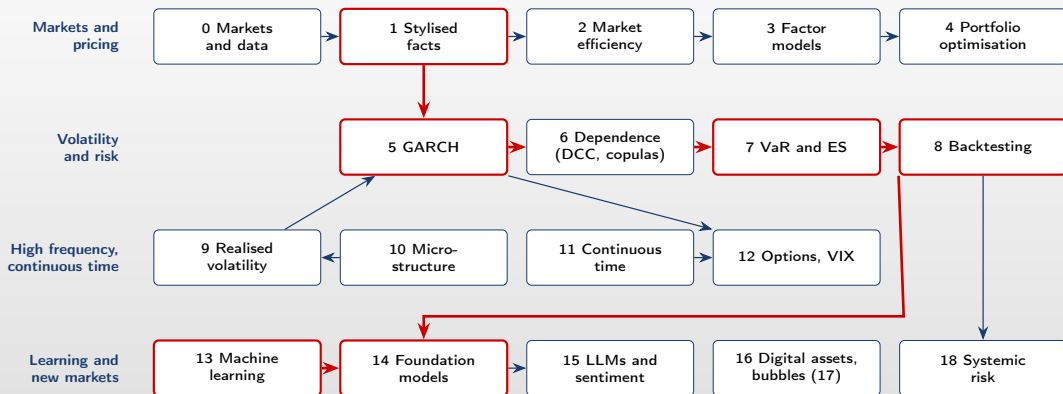
- Team project (GitHub, presentation, oral defence): **70%**
- Quizzes and seminar activity: **20%**
- Attendance: **10%**; there is no final exam



Bucharest University of Economic Studies (ASE), Piața Romană

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## The Course in One Picture



- Red: the main thread of the course, from stylised facts to risk forecasts that we can test
- Blue arrows: where one chapter supplies the input or the benchmark of another

## Reading the Course Map

- Chapter 1 sets the targets: heavy tails, volatility clustering, almost no linear predictability
  - ▶ Every later model is judged by whether it reproduces these facts
- Chapters 5–8 turn the facts into a risk number and a test of that number
  - ▶ GARCH gives  $\sigma_{t+1}$ ; VaR and ES give the loss; backtesting says whether to trust it
- Chapters 9–12 measure volatility better (intraday data, option prices) and in continuous time
- Chapters 13–15 bring flexible models, always evaluated with the tools of Chapter 8
- Chapters 16–18 apply the same toolkit to crypto, bubbles and the whole financial system

## Five Questions for Every Model

### Before estimating

- **Data:** which series, which calendar, which adjustments (dividends, splits, holidays)?
- **Model:** which stylised fact does it capture, and which does it ignore?
- **Estimation:** on which window, with which information at time  $t$ ?

### After estimating

- **Evaluation:** out of sample, with a loss that matches the target (QLIKE, FZ0, Kupiec)?
- **Uncertainty:** robust standard errors, bootstrap, a test against a simple benchmark?
- If one answer is missing, the result is not yet a finding

## Chapter 0: Financial Markets in 2026

### Key idea

- Markets combine exchanges, over-the-counter dealers, exchange-traded funds (ETFs), algorithms and blockchains
- Data quality is a modelling decision: source, adjustment and calendar change the result

### Key formula

- $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$  (compound annual growth rate)
- $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$

### Result from the data

- 2015–2026: SPY (S&P 500 ETF, dividends reinvested) 13.7% a year with 17.6% volatility; Bitcoin 60.6% a year with 66.7%
- BET, July 2007 to February 2009: a drawdown of −82.5%

### Leads to

- Returns, not prices, are the object of study (Chapter 1)

## Chapter 1: Stylised Facts and Returns

### Key idea

- ▣ Returns are uncorrelated but not independent: their size is predictable, their sign almost not
- ▣ Heavy tails and volatility clustering in every market  
Cont (2001)

### Key formula

- ▣  $\hat{K} - 3 = \frac{T^{-1} \sum_t (r_t - \bar{r})^4}{\hat{\sigma}^4} - 3$
- ▣  $Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$

### Result from the data

- ▣ Excess kurtosis from 9.3 (gold) to 17.8 (BET-TR); the Normal distribution gives 0
- ▣ Days beyond  $5\sigma$  are more than 1000 times more frequent than the Normal distribution predicts

### Leads to

- ▣ Volatility models (Chapter 5) and tail risk measures (Chapter 7)

## Chapter 2: Market Efficiency

### Key idea

- Efficiency: excess returns are unpredictable given the information set [Fama \(1970\)](#)
- Efficiency varies over time (adaptive markets); tests need robust inference

### Key formula

- $$VR(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)} = 1$$
 (variance ratio) under a random walk

### Result from the data

- Share of rolling windows that reject the random walk: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- Statistical rejection is not exploitable profit

### Leads to

- Anomalies and factors (Chapter 3); the multiple-testing problem

## Chapter 3: Factor Models and Asset Pricing

### Key idea

- ▣ Expected returns should reward only exposure to systematic factors
- ▣ Hundreds of published factors require higher hurdles Harvey, Liu & Zhu (2016)

### Key formula

- ▣  $E[R_i] - r_f = \beta_i (E[R_m] - r_f)$
- ▣  $\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$

### Result from the data

- ▣ 25 size and book-to-market portfolios: SML slope  $-4.4\%$ ,  $\text{GRS} = 4.18$ : the CAPM is rejected
- ▣ Sector ETFs: the first principal component (the market) explains 66% of the variance

### Leads to

- ▣ Portfolio construction (Chapter 4); factor models inside machine learning (Chapter 13)

## Chapter 4: Portfolio Optimisation

### Key idea

- ▣ Markowitz asks the right question, but estimation error dominates the answer [Markowitz \(1952\)](#)
- ▣ Shrinkage, constraints and risk-based rules stabilise the weights; 1/N is a hard benchmark [DeMiguel, Garlappi & Uppal \(2009\)](#)

### Key formula

- ▣  $w_{\text{tan}} \propto \Sigma^{-1}(\mu - r_f \mathbf{1})$
- ▣  $SR = (\mu - r_f)/\sigma$  (Sharpe ratio)

### Result from the data

- ▣ 60-month samples: median in-sample Sharpe 1.57, median true Sharpe 0.30
- ▣ 96% of the estimated tangency portfolios are truly worse than 1/N

### Leads to

- ▣ Risk budgets need volatility and dependence forecasts (Chapters 5–6)

## Recap: Markets, Returns and Pricing

- ▣ Clean data first: adjusted prices for ETFs and stocks, own calendar for each series, join on prices before returns
- ▣ Stylised facts are the benchmark for every model that follows
- ▣ Efficiency and factor tests need robust errors and multiple-testing corrections
- ▣ Optimisation amplifies estimation error: evaluate out of sample, against  $1/N$

## Chapter 5: Conditional Volatility: GARCH

### Key idea

- ▣ Volatility clusters, so today's shock predicts tomorrow's variance Engle (1982), Bollerslev (1986)
- ▣ Persistence close to 1; asymmetry in equities; heavy-tailed innovations

### Key formula

- ▣  $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
- ▣  $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$

### Result from the data

- ▣ S&P 500, 2000–2026:  $\alpha + \beta = 0.9947$ , a half-life of 131 days
- ▣ GARCH volatility and the VIX: correlation 0.89

### Leads to

- ▣  $\hat{\sigma}_{t+1}$  is the input of filtered VaR and ES (Chapter 7)

## Chapter 6: Multivariate Volatility and Dependence

### Key idea

- ▣ Correlations move with the macro regime; diversification is not guaranteed
- ▣ Copulas separate the margins from dependence; tail dependence drives joint losses

### Key formula

- ▣  $Q_t = (1 - a - b) \bar{Q} + a z_{t-1} z_{t-1}^\top + b Q_{t-1}$
- ▣  $R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$

### Result from the data

- ▣ SPY-TLT (long US Treasury bond ETF):  $\rho$  from  $-0.40$  to  $0.11$  after 2022, so a 60/40 portfolio went from 10.4% to 13.2% volatility
- ▣ Bitcoin-SPY DCC correlation: 0.04 before 2020, 0.31 since

### Leads to

- ▣ Portfolio VaR and Euler contributions (Chapter 7); systemic risk (Chapter 18)

## Chapter 7: Value-at-Risk and Expected Shortfall

### Key idea

- ▣ VaR is a quantile of the loss, ES the average loss beyond it; ES is coherent
- ▣ Filtered historical simulation and conditional EVT react to volatility in days McNeil & Frey (2000)

### Key formula

- ▣  $\text{VaR}_\alpha(X) = -q_\alpha(X)$
- ▣  $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$

### Result from the data

- ▣ S&P 500, daily: historical  $\text{VaR}_{1\%} = 3.45\%$ ,  $\text{ES}_{1\%} = 5.01\%$
- ▣ The Normal distribution gives only 2.80%: heavy tails matter

### Leads to

- ▣ A risk forecast is useful only if it survives a backtest (Chapter 8)

## Chapter 8: Backtesting and Evaluating Risk Forecasts

### Key idea

- Test the number and the timing of breaches: coverage and independence [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- Rank (VaR, ES) forecasts with a consistent score (FZ0) and Diebold–Mariano tests [Fissler & Ziegel \(2016\)](#)

### Key formula

- $LR_{uc} = -2 \ln \frac{(1 - \alpha)^{T-x} \alpha^x}{(1 - \hat{\pi})^{T-x} \hat{\pi}^x} \sim \chi_1^2$
- $\hat{\pi} = x/T$

### Result from the data

- S&P 500, VaR 1%, 2007–2026: Normal 2.70%, HS 1.67%, FHS 1.23%, GARCH-EVT 1.13% breaches
- A GARCH filter plus a flexible tail wins in all four markets

### Leads to

- The same tests judge realised-volatility and machine-learning forecasts (Chapters 9, 13, 14)

## Recap: Volatility, Dependence and Risk

- ▣ GARCH turns volatility clustering into one-day-ahead forecasts
- ▣ Dependence is dynamic and strongest in the tails; crisis correlations need a volatility adjustment
- ▣ VaR and ES must combine a volatility filter with a heavy-tailed distribution
- ▣ A risk model is accepted only after tests of coverage, independence and ES

## Chapter 9: Realised Volatility

### Key idea

- ▣ Intraday prices measure daily variance ( $RV$ , realised variance); the log-HAR (heterogeneous autoregressive) model captures its long memory [Corsi \(2009\)](#)
- ▣ Returns divided by realised volatility are close to the Normal distribution

### Key formula

- ▣  $RV_t = \sum_{i=1}^M r_{t,i}^2$
- ▣  $\ln RV_{t+1} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln RV_t^{(w)} + \beta_m \ln RV_t^{(m)} + \varepsilon_{t+1}$

### Result from the data

- ▣ SPY: kurtosis 15.73 for open-to-close returns, 2.65 after dividing by  $\sqrt{RV_t}$
- ▣ log-HAR beats GARCH(1,1)- $t$  in QLIKE by 10% for SPY ( $p = 0.021$ ) and 22% for Bitcoin

### Leads to

- ▣ Better volatility inputs for VaR; microstructure explains the noise (Chapter 10)

## Chapter 10: Market Microstructure

### Key idea

- Prices form in limit order books; the spread pays for processing, inventory and adverse selection
- Liquidity follows the clock and the VIX, and disappears in stress

### Key formula

- $ILLIQ = \frac{1}{D} \sum_{d=1}^D \frac{|r_d|}{DVOL_d}$  (Amihud illiquidity;  $DVOL_d$ : traded value on day  $d$ )
- $s_{Roll} = 2\sqrt{-Cov(\Delta p_t, \Delta p_{t-1})}$  (Roll spread)

### Result from the data

- Amihud ratio (bp per USD 1 million): SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
- SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar

### Leads to

- Liquidity risk in VaR and in the project data choices

## Chapter 11: Continuous-Time Models

### Key idea

- Brownian motion is the limit of random walks; Itô's lemma solves the price equations
- Jumps explain the tails, stochastic volatility the clustering

### Key formula

- $dS_t = \mu S_t dt + \sigma S_t dW_t \Rightarrow S_T = S_0 \exp\left((\mu - \frac{\sigma^2}{2})T + \sigma W_T\right)$

### Result from the data

- S&P 500 under geometric Brownian motion: excess kurtosis 10.9 instead of 0
- 62 days beyond  $\pm 4$  standard deviations, against 0.59 expected

### Leads to

- Option pricing needs only  $\sigma$ , not  $\mu$  (Chapter 12)

## Chapter 12: Options and the Volatility Surface

### Key idea

- ▣ Black–Scholes prices options by hedging; implied volatility is the market's view of  $\sigma$  **Black & Scholes (1973)**
- ▣ Smile and skew encode the fear of crashes

### Key formula

- ▣  $C = S N(d_1) - Ke^{-rT} N(d_2)$
- ▣  $VRP_t = VIX_t^2 - RV_{t,t+21}$

### Result from the data

- ▣ 1990–2026: mean VIX 19.4, mean realised volatility over the next month 15.3
- ▣ The variance risk premium is positive on 86% of days, but not in crashes

### Leads to

- ▣ Implied volatility as a feature for machine learning and as a stress indicator (Chapters 13, 18)

## Recap: High Frequency, Continuous Time and Options

- ▣ Intraday data measure volatility; log-HAR forecasts it better than daily GARCH
- ▣ Liquidity is a risk of its own and varies by orders of magnitude across markets
- ▣ Continuous-time models are tractable, but realistic ones need jumps and random volatility
- ▣ Option prices carry forward-looking information, including a variance risk premium

## Chapter 13: Machine Learning in Finance

### Key idea

- Flexible models, but little signal and much noise; the finance-specific part is the process [López de Prado \(2018\)](#)
- Purged cross-validation, meta-labelling and deflated Sharpe ratios [Bailey & López de Prado \(2014\)](#)

### Key formula

- $(1 - B)^d x_t = \sum_{k \geq 0} w_k x_{t-k}$
- $w_0 = 1, w_k = -w_{k-1} \frac{d - k + 1}{k}$

### Result from the data

- Shuffled K-fold reports 69.4% accuracy on pure noise: leakage
- S&P 500: fractional differencing is stationary at  $d^* = 0.25$ ; no model beats “always up”

### Leads to

- Deep learning and foundation models, judged the same way (Chapter 14)

## Chapter 14: Deep Learning and Time-Series Foundation Models

### Key idea

- ▣ LSTMs (long short-term memory networks) and Transformers learn from sequences; foundation models forecast zero-shot
- ▣ Evaluate them with Chapter 8 tools, after the release of their weights

### Key formula

- ▣  $L(RV, h) = \frac{RV}{h} - \ln \frac{RV}{h} - 1$  (QLIKE loss)

### Result from the data

- ▣ SPY realised volatility, QLIKE: Chronos-2 0.231, log-HAR 0.234, GARCH- $t$  0.283; the first two are not significantly different
- ▣ Raw Chronos-2 VaR 1% on the S&P 500: 1.67% breaches

### Leads to

- ▣ Text as data (Chapter 15)

## Chapter 15: LLMs and Sentiment Analysis

### Key idea

- From dictionaries to FinBERT and large language models (LLMs): text becomes a number
- Text embeddings enter forecasting models as features

### Key formula

- $S_t = \frac{N_t^+ - N_t^-}{N_t^+ + N_t^-}$ , with  $N_t^\pm$  the number of positive and negative texts on day  $t$
- $r_{t+1} = a + b S_t + \varepsilon_{t+1}$  (predictive regression)

### Questions the chapter answers

- Does a sentiment score predict returns or volatility out of sample, after costs?
- Was the text available at time  $t$ , or does it leak future information?

### Leads to

- Sentiment in crypto markets and bubbles (Chapters 16–17)

## Chapter 16: Digital Assets and DeFi

### Key idea

- Cryptocurrencies are a statistically distinct asset class, mainly through their heavy tails [Pele et al. \(2023\)](#)
- Stablecoins, automated market makers and spot ETFs connect crypto to mainstream finance; decentralised finance (DeFi) runs on smart contracts

### Key formula

- $x \cdot y = k$  (constant-product market maker;  $x, y$ : the two token reserves)
- $\text{Depeg} = |P_t - 1|$ , with  $P_t$  the stablecoin price in USD

### Questions the chapter answers

- How often and how far do stablecoins leave their peg?
- Did spot ETFs change the risk and the correlations of Bitcoin?

### Leads to

- Explosive price episodes (Chapter 17)

## Chapter 17: Bubbles and Crashes

### Key idea

- ▣ A bubble is a price path that grows faster than fundamentals can justify
- ▣ Right-tailed unit-root tests date the start and the end of explosive episodes

### Key formula

- ▣  $GSADF = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2}$
- ▣  $ADF_{r_1}^{r_2}$ : the ADF statistic on the sample fraction  $[r_1, r_2]$ ;  $r_0$ : minimum window

### Questions the chapter answers

- ▣ Which crypto or technology price episodes were explosive, and when did they end?
- ▣ Can a log-periodic model predict the crash date, or only fit it afterwards?

### Leads to

- ▣ Contagion from one market to the system (Chapter 18)

## Chapter 18: Systemic Risk and Networks

### Key idea

- Systemic risk: losses of one institution that spread to the whole system
- CoVaR (the VaR of the system conditional on the distress of one institution), SRISK and spillover networks measure the contribution of each institution

### Key formula

- $$\Delta \text{CoVaR}^i = \text{CoVaR}_{\alpha}^{\text{sys}|X^i = -\text{VaR}_{\alpha}^i} - \text{CoVaR}_{\alpha}^{\text{sys}|X^i = \text{median}}$$
- CoVaR: the VaR of the system return, conditional on the state of institution  $i$

### Questions the chapter answers

- Which banks contribute most to the tail risk of the system, and does it change in crises?
- How connected are Romanian and euro-area markets in stress?

### Leads to

- Stress testing and the team projects

## Recap: Learning, Text and New Markets

- ▣ Flexible models need a strict process: purging, embargo, deflated performance measures
- ▣ Foundation models tie with strong benchmarks on volatility and do not replace GARCH-based FHS for tail risk
- ▣ Text, crypto, bubbles and systemic risk reuse the same statistical toolkit
- ▣ New data and new models: the same five questions apply

## Six Pitfalls That Returned in Every Chapter

### Pitfalls of design

- ▣ **Look-ahead bias:** information from after  $t$  used at  $t$
- ▣ **Overfitting:** a model tuned to the sample it is judged on
- ▣ **Multiple testing:** the best of many tries reported as if it were the only one

### Pitfalls of data and inference

- ▣ **Wrong calendar join:** returns computed before aligning assets
- ▣ **Heavy tails ignored:** standard errors that assume the Normal distribution and independence
- ▣ **Wrong loss:** a loss function that cannot rank the forecast target

## Pitfall 1: Look-Ahead Bias

- Any parameter, threshold or scaling estimated on the full sample leaks the future
  - ▶ Examples: full-sample GARCH parameters, full-sample standardisation, labels that overlap the test set
- BET, VaR 1%, 2005–2026: GARCH- $t$  with full-sample parameters breaches on 1.31% of days, with rolling estimation on 1.47%
  - ▶ The look-ahead version looks better than any forecaster could have done in real time
- Chapter 13: shuffled K-fold reports 69.4% accuracy on pure noise
- Remedy: rolling or expanding windows; purging and embargo; write the information set  $\mathcal{F}_{t-1}$  next to every forecast

## Pitfall 2: Overfitting

- The in-sample fit rewards noise that will not repeat
  - ▶ Chapter 4: in-sample Sharpe 1.57, true Sharpe 0.30 for the same portfolios
- Backtest overfitting: trying many strategies and keeping the best [Bailey et al. \(2017\)](#)
  - ▶ The deflated Sharpe ratio corrects for the number of trials and for non-normal returns [Bailey & López de Prado \(2014\)](#)
- Remedy: fix the design before looking at the test period; report every model you tried; compare with a simple benchmark (1/N, historical simulation, random walk, log-HAR)

## Pitfall 3: Multiple Testing

- Chapter 3: with 300 useless factors, on average 15.0 pass  $|t| > 1.96$ , and the best reaches  $|t| \approx 3.06$ 
  - ▶ This is why the factor literature asks for  $|t| > 3$  [Harvey, Liu & Zhu \(2016\)](#)
- Calendar anomalies, trading rules, model specifications: each extra test adds a chance of a false discovery
- Remedy: count your tests and correct
  - ▶ Bonferroni and Holm control the chance of any false discovery; Benjamini–Hochberg controls their share
  - ▶ For forecasts: the model confidence set instead of many pairwise tests

## Pitfall 4: Joining on Returns Instead of Prices

- ❑ Bitcoin trades every day, SPY only on weekdays
  - ▶ Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
  - ▶ Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- ❑ Bitcoin, 2020–2026, on the common days: cumulative return 1025% (right) against 680% (wrong)
  - ▶ Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong join hides weekend risk
- ❑ Remedy: align prices on common days first; one series alone stays on its own calendar

## Pitfall 5: Heavy Tails Ignored in Inference

- ▣ Classical standard errors assume independence and finite fourth moments
  - ▶ BET excess kurtosis 11.1, with a block-bootstrap 95% CI of [8.0, 13.9]: wide even with 6 670 days
- ▣ BET GARCH persistence: 95% CI [0.974, 1.007], so the half-life lies between 26 days and infinity
  - ▶ A point estimate of the half-life says little on its own
- ▣ Remedy: robust (HAC) errors, block bootstrap, tests designed for heavy tails; report intervals, not only points

## Pitfall 6: A Loss Function That Cannot Rank the Target

- Variance forecasts are compared with a noisy proxy ( $r_t^2$  or  $RV_t$ )
  - ▶ MSE and QLIKE rank them consistently; QLIKE is more robust to extreme days
  - ▶ Chapter 9: log-HAR beats GARCH in QLIKE; the MSE comparison is inconclusive
- ES alone cannot be ranked by any scoring function; the pair (VaR, ES) can [Fissler & Ziegel \(2016\)](#)
  - ▶ Use FZ0 with Diebold–Mariano tests [Diebold & Mariano \(1995\)](#); expect finite-sample limits on precision [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- Remedy: choose the loss before the comparison, and state it in the project report

## Recap: A Checklist Against the Pitfalls

### Design

- ▣ Every forecast uses only  $\mathcal{F}_{t-1}$
- ▣ Test period fixed in advance and used once
- ▣ All models tried are reported; tests are counted

### Data and inference

- ▣ Prices joined before returns; adjustments and calendars documented
- ▣ Robust or bootstrap uncertainty for every headline number
- ▣ Loss function consistent for the target; comparison with a simple benchmark

## One Question, Four Chapters

### The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

### Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6670 daily log returns
- Days with an unchanged close (holidays filled with the previous value) removed

### The four steps

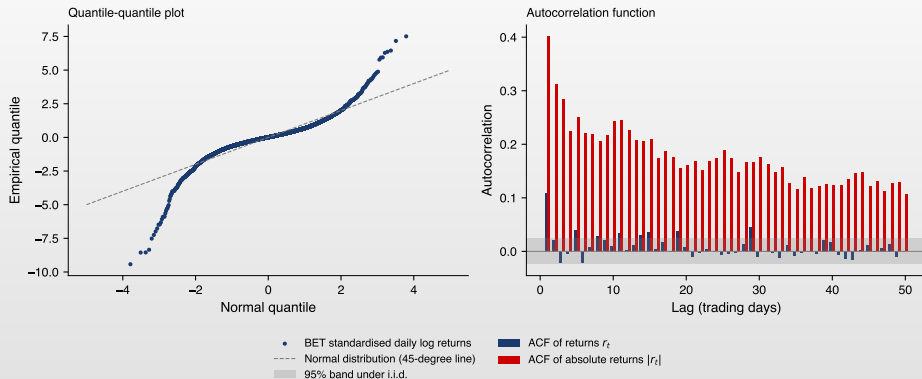
- Chapter 1: stylised facts, with robust inference
- Chapter 5: GARCH(1,1) with Student- $t$  innovations
- Chapter 7: one-day VaR 1% from four models
- Chapter 8: coverage, independence and Basel zones



The Stock Exchange Palace, Bucharest (2023)

Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

## Step 1: Stylised Facts of the BET

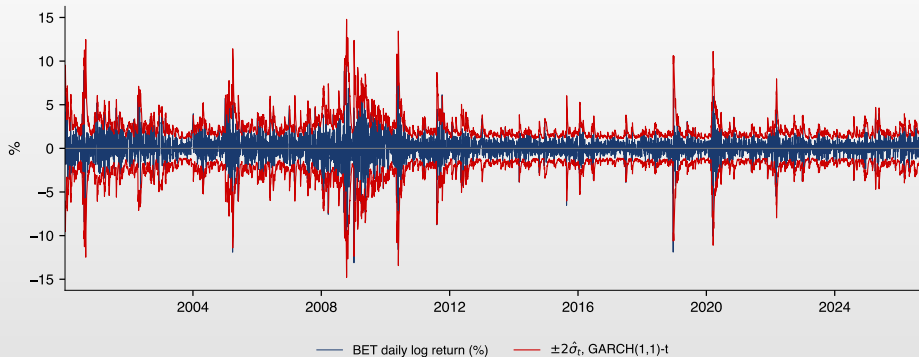


□ Left: quantiles of standardised returns against the Normal distribution; right: autocorrelations of  $r_t$  and  $|r_t|$ , lags 1–50

## Reading Step 1

- ▣ Heavy tails: excess kurtosis 11.1 (block-bootstrap 95% CI [8.0, 13.9]); skewness  $-0.65$ 
  - ▶ 51 days beyond  $\pm 4$  standard deviations, against 0.42 expected under the Normal distribution
  - ▶ Worst day:  $-13.1\%$  (log) on 7 Jan 2009
- ▣ Clustering: first-order autocorrelation of  $|r_t|$  is 0.40 and decays slowly; Ljung–Box  $Q(10)$  on  $r_t^2$  is 2440
  - ▶ Returns themselves have first-order autocorrelation 0.11: the BET is less liquid than large markets (Chapters 2, 10)
- ▣ Consequence: a Normal, constant-volatility VaR will be too low in stress and too high in calm periods

## Step 2: GARCH(1,1)- $t$ Volatility



- BET daily log returns with  $\pm 2\hat{\sigma}_t$  from a GARCH(1,1) with Student- $t$  innovations, estimated on 2000–2026

## Reading Step 2

- ▣ Estimates (robust standard errors):  $\hat{\alpha} = 0.192$  (0.025),  $\hat{\beta} = 0.799$  (0.024),  $\hat{\nu} = 5.01$ 
  - ▶  $\hat{\nu} \approx 5$ : even after filtering, shocks have heavy tails
- ▣ Persistence  $\hat{\alpha} + \hat{\beta} = 0.9907$ : a half-life of about 74 trading days
  - ▶ But the 95% CI of the persistence includes 1 (Pitfall 5)
- ▣ Peak daily volatility 7.39% on 15 Oct 2008; at the end of the sample 1.67%

## Step 3: Four VaR 1% Forecasts

### Unconditional (last 500 days)

- ▣ **HS**: the loss  $L = -r$  exceeded on 1% of the last 500 days (empirical quantile)
- ▣ **Normal**:  $\text{VaR} = -(\hat{\mu} + \hat{\sigma} z_{0.01})$ ,  $z_{0.01} = -2.326$

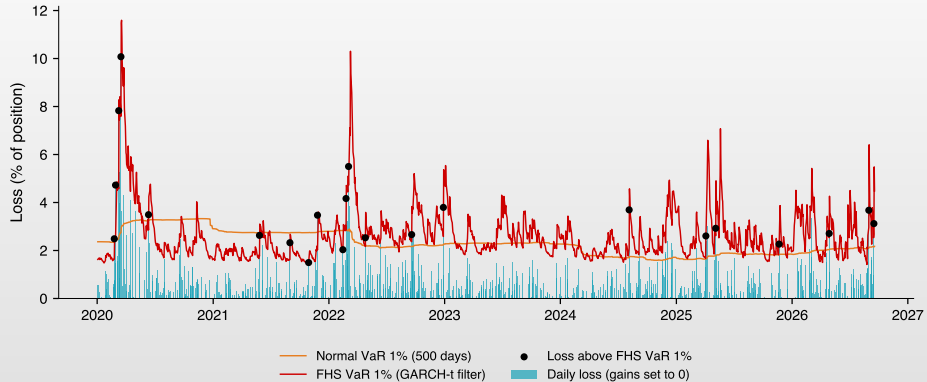
### Conditional (GARCH- $t$ on the last 1000 days)

- ▣ **GARCH- $t$** :  $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t t_\nu^{-1}(0.01) \sqrt{(\nu - 2)/\nu})$
- ▣ **FHS**:  $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \hat{q}_{0.01}(z))$ , the 1% quantile of the standardised residuals [McNeil & Frey \(2000\)](#)

### Real-time design

- ▣ Forecast for day  $t$  uses data up to  $t - 1$  only
- ▣ GARCH re-estimated every 21 trading days;  $\hat{\sigma}_t$  updated daily
- ▣ Evaluation: 3 Jan 2005 – 18 Sep 2026,  $T = 5\,432$  days
- ▣ Expected breaches at 1%: about 54

### Step 3: Losses and VaR 1%, 2020–2026

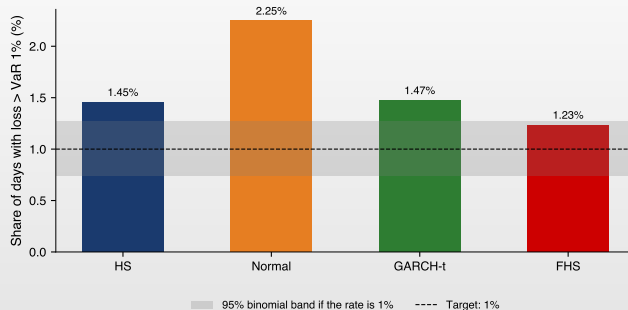


□ Daily BET losses (gains set to 0), Normal and FHS VaR 1%; black dots: losses above the FHS VaR

## Reading Step 3

- FHS reacts within days: it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash
  - ▶ The Normal VaR on a 500-day window moves only slowly
- 2020–2026 (1 675 days, about 17 breaches expected): FHS 22 breaches, Normal 41
  - ▶ Normal breaches cluster in the first weeks of every stress episode
- Last forecast in the sample: FHS 4.48%, Normal 2.18%: the two models disagree most when volatility changes

## Step 4: Breach Rates



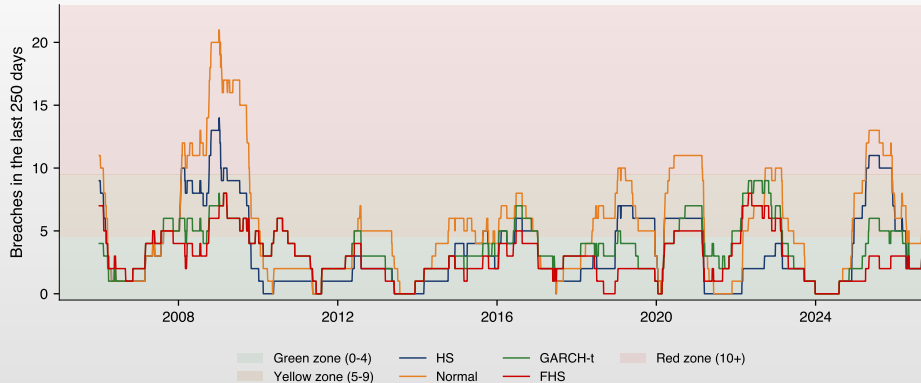
- Share of days with a loss above the VaR 1% forecast, 2005–2026; grey band: 95% binomial range if the true rate is 1%

## Reading Step 4: Coverage and Independence

| Model   | Breaches | Rate  | $p_{uc}$ | $p_{ind}$ | $p_{cc}$ | Red zone |
|---------|----------|-------|----------|-----------|----------|----------|
| HS      | 79       | 1.45% | 0.002    | <0.001    | <0.001   | 5.5%     |
| Normal  | 122      | 2.25% | <0.001   | <0.001    | <0.001   | 18.6%    |
| GARCH-t | 80       | 1.47% | 0.001    | 0.864     | 0.005    | 0.0%     |
| FHS     | 67       | 1.23% | 0.095    | 0.267     | 0.134    | 0.0%     |

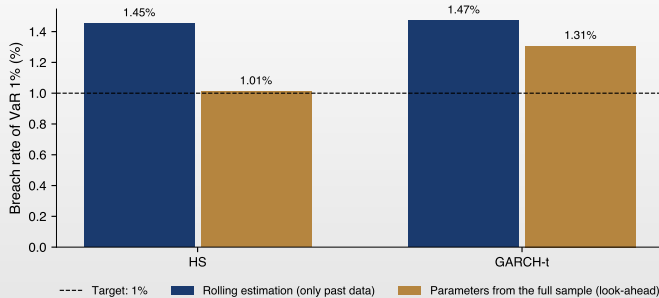
- ▣  $p_{uc}$ : Kupiec coverage test;  $p_{ind}$ : Christoffersen independence test;  $p_{cc}$ : both jointly [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- ▣ Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- ▣ Only FHS passes all three tests at 5%; HS and Normal also fail independence: their breaches cluster
- ▣ GARCH- $t$  has independent breaches but too many of them: the Student- $t$  tail is not heavy enough at 1%

## Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- The Normal model spends 18.6% of the windows in the red zone, HS 5.5%, the two GARCH-filtered models never

## The Look-Ahead Trap in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction; full-sample GARCH-t: 1.31%
- Both look better than their real-time versions: the gap is the value of information nobody had

## Recap: The Case Study

- ▣ The BET shows every stylised fact: heavy tails, clustering, some linear autocorrelation
- ▣ GARCH- $t$  captures clustering; the filtered residuals still have heavy tails
- ▣ A GARCH filter with an empirical tail (FHS) is the only VaR 1% that passes coverage and independence
- ▣ Real-time design matters: full-sample estimates flatter every model
- ▣ The same four steps are a template for your project

## What the Project Is

- ▣ An empirical study of one question about financial markets, answered with the methods of this course
  - ▶ Teams; four steps: replicate, extend, deliver on GitHub, present and defend (next slide)
- ▣ It carries **70%** of the final grade; quizzes and seminar activity carry 20%, attendance 10%
- ▣ Starting points: the Part C question of every seminar (next section)
  - ▶ Your own question is welcome if it passes the five questions of this chapter
- ▣ A good project can end with a negative answer, if the evidence is honest and precise

## The Project in Four Steps

- ▣ **1. Replicate:** reproduce one published number (a table, a coefficient, a backtest result) from a paper or from the lecture
  - ▶ An AI assistant cannot guess an exact published figure: matching it shows that your data and code are right
- ▣ **2. Extend:** new data, a new market (BVB, crypto) or a more recent sample
  - ▶ One clear question, answered with a test, not only with a chart
- ▣ **3. Deliver on GitHub:** code, data description and a README that reproduces every result
  - ▶ `AI_USE.md` and `AI_ERRORS.md` are mandatory
- ▣ **4. Present and defend:** team presentation, then a 10-minute oral defence
  - ▶ Each member explains one result and answers a “what changes if...” question; grades can differ between members

## Choosing the Question

### A good question is

- **Specific:** one market, one period, one quantity
- **Testable:** a null hypothesis and a benchmark
- **Feasible:** data you can obtain and clean in a few weeks
- **Relevant:** an answer matters to an investor, a bank or a regulator

### Examples

- Too vague: "Is Bitcoin risky?"
  - ▶ Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin, and which model forecasts it best since 2024?"
- Too broad: "Can machine learning predict the market?"
  - ▶ Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

## Data

- Use the course data where possible: daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
  - ▶ Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
  - ▶ Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
  - ▶ Remove holiday fills and weekend quotes where the market is closed; annualise with the real frequency
  - ▶ Several assets: join on prices, then compute returns
- Look for bad prints before modelling: jumps of  $\pm 50\%$  in adjusted prices are usually data errors

## Methods: From Question to Chapter

| If your question is about                                                           | Start from                                                                                                                   |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Return distribution, tails, predictability<br>Expected returns, factors, portfolios | Chapters 1–2: moments, robust tests, variance ratios<br>Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample<br>Sharpe tests |
| Volatility and correlation<br>Risk measures and capital                             | Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR<br>Chapters 7–8: FHS, EVT, backtests, FZ0                                     |
| Liquidity, options, continuous time                                                 | Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk<br>premium                                                         |
| Forecasting with ML, deep learning or text<br>Crypto, bubbles, systemic risk        | Chapters 13–15: purged CV, foundation models, sentiment<br>Chapters 16–18: stablecoins, GSADF, CoVaR, networks               |

## Honest Evaluation Checklist

### Before you look at the results

- ☐ Write the hypothesis, the benchmark and the loss function
- ☐ Fix the estimation window and the test period
- ☐ Decide how many models and parameters you will try

### When you report

- ☐ Out-of-sample results only for claims about forecasting
- ☐ Every headline number with a standard error, CI or  $p$ -value
- ☐ Costs for trading strategies; subperiods and a second market for robustness
- ☐ What did **not** work, and why

## The Report

- ▣ **1. Question and motivation:** one paragraph, one hypothesis
- ▣ **2. Literature:** three to eight clickable references, including the course textbooks [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#)
- ▣ **3. Data and replication:** sources, period, cleaning rules, descriptive statistics; the published number and your value
- ▣ **4. Methods:** models, estimation window, benchmark, loss function, tests
- ▣ **5. Results:** tables and charts, each with an interpretation
- ▣ **6. Robustness and limitations; 7. Conclusion:** the answer, in one sentence
- ▣ **Code:** a GitHub repository whose README and notebook reproduce every number, with [AI\\_USE.md](#) and [AI\\_ERRORS.md](#)

## The Presentation

### Structure (about 12 minutes)

- ▣ The question and why it matters (1 slide)
- ▣ Data and one chart that shows the problem (2 slides)
- ▣ Method and benchmark (2 slides)
- ▣ Main result with its uncertainty (2–3 slides)
- ▣ Robustness, limitations, answer (2 slides)

### Good practice

- ▣ One message per slide; charts with readable axes and the legend below
- ▣ Every team member presents and can answer questions on the code
- ▣ Then a 10-minute oral defence: each member explains one result and answers a “what changes if...” question, without AI

## Project Grading (70% of the Final Grade)

| Component                              | What we look for                                                                    |
|----------------------------------------|-------------------------------------------------------------------------------------|
| Replication (step 1)                   | A published number reproduced; any gap explained                                    |
| Question, data and extension (step 2)  | Specific, testable question; documented data and calendar alignment                 |
| Methods and inference                  | Real-time design, a benchmark, out-of-sample tests, uncertainty                     |
| Interpretation                         | Economic meaning, limitations, honest conclusion                                    |
| GitHub repository (step 3)             | README that reproduces every result; AI_USE.md, AI_ERRORS.md                        |
| Presentation and oral defence (step 4) | Individual: one result explained, answers to “what changes if...”                   |
| <b>Total</b>                           | <b>70%</b> of the final grade; plus quizzes and activity (20%) and attendance (10%) |

- Grades can differ between team members, depending on the oral defence

## Timeline

- Week 1 after this chapter: team and question
  - ▶ Submit a one-page proposal (template in Seminar 19, Part C); open the GitHub repository
- Weeks 2–3: data and the replication of one published number (step 1)
- Weeks 4–5: extension, evaluation, robustness (step 2); feedback at the seminar
- Final weeks: repository with README and the two AI files (step 3); presentation and oral defence (step 4)
- Exact dates are announced at the course and seminar

## Where Projects Usually Lose Points

- ▣ A question that changes after the results are known
- ▣ In-sample fit presented as forecasting power
- ▣ No benchmark, or a weak one chosen to be beaten
- ▣ Numbers without uncertainty; many tests without correction
- ▣ Charts without units, legends inside the data, unexplained acronyms
- ▣ Code that does not run, or numbers in the report that the code does not produce
- ▣ AI output kept without checks: undeclared use, unchecked code, references that do not exist

## Recap: The Team Project

- ▣ One specific, testable question; one benchmark; one loss function chosen in advance
- ▣ Clean data with the course conventions; real-time design; honest reporting
- ▣ Replicate, extend, deliver on GitHub, present and defend: 70% of the grade

## The AI Policy of the Course

- ▣ **Allowed:** AI assistants for code, debugging, literature search and writing, in the project and in the seminars
- ▣ **Declared:** every tool, what it was used for and what you checked yourselves, in [AI\\_USE.md](#)
  - ▶ Undeclared use of AI counts as plagiarism
- ▣ **Checked:** you are responsible for every line of code and text
  - ▶ “The AI wrote it” is not an explanation at the oral defence
  - ▶ Every reference needs a working DOI or link; a reference that does not exist counts as fabricated data
- ▣ **No AI during the oral defence:** you answer alone, from your own understanding of the project

## The Two Mandatory Files

### AI\_USE.md: the declaration

- ▣ Tools: name, version, who used them and for what
- ▣ Files or sections where AI-written code or text was kept, and what you changed
- ▣ How you verified the replicated number, the data, the formulas and the references
- ▣ What the team did without AI

▣ Templates in English and Romanian: [github.com/danpele/MFM/tree/main/project](https://github.com/danpele/MFM/tree/main/project)

### AI\_ERRORS.md: the error log

- ▣ At least **three** places where the assistant was wrong
- ▣ For each: what you asked, what it answered, what was wrong, how you found out, the correct version
- ▣ Errors you caught are a strength of the project, not a weakness

## Typical AI Errors in Financial Modelling (1/2)

- ▣ **Look-ahead bias:** a rolling statistic or a scaler that includes day  $t$  in the forecast for day  $t$ 
  - ▶ Check: change the last observation; the forecast for that day must not move; shift by one day (`.shift(1)`)
- ▣ **Sign of VaR:** the 1% quantile of returns reported as VaR, or the 99% quantile of returns (a gain) used as the loss
  - ▶ Check:  $\text{VaR}_\alpha = -q_\alpha(r) > 0$  for a loss; VaR 1% of a daily index is a few percent, not negative
- ▣ **Annualisation:** daily volatility multiplied by 252 instead of  $\sqrt{252}$ , or 252 days used for crypto that trades every day
  - ▶ Check: means scale with  $h$ , volatilities with  $\sqrt{h}$ ; use the real number of trading days (365 for crypto)

## Typical AI Errors in Financial Modelling (2/2)

- ❑ **Random cross-validation (CV) on time series:** shuffled K-fold trains on the future; in Chapter 13 it reports 69.4% accuracy on pure noise
  - ▶ Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo [López de Prado \(2018\)](#)
- ❑ **Invented references:** plausible authors, title and journal, but the paper does not exist
  - ▶ Check: search every DOI on [Crossref](#); open the paper and find the claim you cite
- ❑ **Also frequent:** wrong degrees of freedom in a test, library defaults you did not choose, numbers stated without running any code
  - ▶ Every seminar has a C2 exercise that trains you to catch such errors

## Recap: Using AI in Your Project

- Use AI freely, declare it in `AI_USE.md`, and check every line it writes
- The replicated number is your best test: an AI cannot guess it
- Log at least three AI errors in `AI_ERRORS.md`; at the oral defence you answer alone

## Project Seeds: Part C of Every Seminar (1/2)

| Ch. | Open question (link to the seminar)                                                    |
|-----|----------------------------------------------------------------------------------------|
| 0   | <a href="#">Did spot ETFs change Bitcoin's correlation with equities?</a>              |
| 1   | <a href="#">Did BET-TR autocorrelation fall after the FTSE reclassification?</a>       |
| 2   | <a href="#">Did the BVB become more efficient after the 2020 FTSE upgrade?</a>         |
| 3   | <a href="#">Do factor premia exist on the Bucharest Stock Exchange?</a>                |
| 4   | <a href="#">An optimal portfolio of BVB blue chips in RON</a>                          |
| 5   | <a href="#">Is the Romanian market's volatility driven by the US?</a>                  |
| 6   | <a href="#">Did Romanian and euro-area equities become more integrated after 2020?</a> |
| 7   | <a href="#">Capital for a BVB portfolio vs an S&amp;P 500 portfolio</a>                |

- Each seminar gives the data, a starting method and a reference analysis discussed in class

## Project Seeds: Part C of Every Seminar (2/2)

| Ch. | Open question (link to the seminar)                                                    |
|-----|----------------------------------------------------------------------------------------|
| 8   | <a href="#">Which risk model would you certify for a Romanian bank's trading book?</a> |
| 9   | <a href="#">Is Bitcoin's volatility more predictable than the S&amp;P 500's?</a>       |
| 10  | <a href="#">Does illiquidity carry a premium on the BVB?</a>                           |
| 11  | <a href="#">Which continuous-time model fits BET and Bitcoin?</a>                      |
| 12  | <a href="#">Is the variance risk premium a tradable signal?</a>                        |
| 13  | <a href="#">Does a foundation model beat boosting?</a>                                 |
| 14  | <a href="#">Can a zero-shot foundation model replace GARCH for the BET?</a>            |

- ▣ Seminars 15–18 add questions on sentiment, digital assets, bubbles and systemic risk

## Turning a Seed into a Project

- Replicate first: one published number from a paper behind the seed (step 1)
- Narrow it: one market, one period, one number to forecast or test
  - ▶ Example: Seminar 8 becomes “Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 BET–EUR portfolio in 2015–2026?”
- Extend it: a second market, a new model, a longer or more recent sample
  - ▶ Example: add a foundation model (Chapter 14) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

## Key Takeaways

- ▣ Returns are hard to predict; their risk is not: volatility, tails and dependence can be forecast
- ▣ Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- ▣ Heavy tails change inference: use robust errors and the bootstrap, report intervals
- ▣ Most wrong results come from the design, not from the model: look-ahead, overfitting, multiple testing, calendars
- ▣ New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

## Further Study

### Textbooks

- Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)

### Advanced courses on quantinar.com

-  Statistics of Financial Markets
-  Measuring Statistical Risk

### Code and data

- Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_19](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_19)

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